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Studierichting: Informatica
Oefeningenreeks 3D nr. 7.2

Bepaal de extrema van $f(x) = e^x \sin x$.

1. Eerste afgeleide

$$\begin{aligned} f'(x) &= \frac{d}{dx}(e^x \sin x) \\ &= \left(\frac{d}{dx} e^x\right) \sin x + e^x \left(\frac{d}{dx} \sin x\right) \\ &= e^x \sin x + e^x \cos x \\ &= e^x (\sin x + \cos x) \end{aligned}$$

2. Kritische punten

We zoeken $f'(x) = 0$.

$$e^x (\sin x + \cos x) = 0$$

Omdat $e^x > 0$ voor alle x geldt:

$$\sin x + \cos x = 0$$

$$\Leftrightarrow \sin x = -\cos x$$

$$\Leftrightarrow \tan x = -1$$

$$\Leftrightarrow x_k = -\frac{\pi}{4} + k\pi, \quad k \in \mathbb{Z}$$

3. Tweede afgeleide

$$\begin{aligned} f''(x) &= \frac{d}{dx}(e^x(\sin x + \cos x)) \\ &= \left(\frac{d}{dx} e^x\right)(\sin x + \cos x) + e^x \left(\frac{d}{dx}(\sin x + \cos x)\right) \\ &= e^x(\sin x + \cos x) + e^x(\cos x - \sin x) \\ &= 2e^x \cos x \end{aligned}$$

Daarom:

$$f''(x_k) = 2e^{x_k} \cos x_k = 2e^{x_k} \frac{\sqrt{2}}{2} (-1)^k = \sqrt{2} e^{x_k} (-1)^k$$

4. Classificatie en specificatie

Uit $x_k = -\frac{\pi}{4} + k\pi$ onderscheiden we:

- Stel $k = 2n$ (even).

$$x_{2n} = -\frac{\pi}{4} + 2n\pi.$$

Dan $(-1)^{2n} = +1$, dus $f''(x_{2n}) > 0$.

$\Rightarrow$ lokaal minimum bij

$$x = -\frac{\pi}{4} + 2n\pi, \quad n \in \mathbb{Z}$$

- Stel $k = 2n + 1$ (oneven).

$$x_{2n+1} = -\frac{\pi}{4} + (2n+1)\pi = -\frac{\pi}{4} + 2n\pi + \pi = \frac{3\pi}{4} + 2n\pi.$$

Dan $(-1)^{2n+1} = -1$, dus $f''(x_{2n+1}) < 0$.

$\Rightarrow$ lokaal maximum bij

$$x = \frac{3\pi}{4} + 2n\pi, \quad n \in \mathbb{Z}$$

5. Waarden in de extrema

$$f(x_k) = e^{x_k} \sin x_k = e^{x_k} \left(-\frac{\sqrt{2}}{2} (-1)^k \right) = -\frac{\sqrt{2}}{2} (-1)^k e^{x_k}$$

- Lokaal minimum ($k = 2n$):

$$x = -\frac{\pi}{4} + 2n\pi, \quad f(x) = -\frac{\sqrt{2}}{2} e^{-\frac{\pi}{4} + 2n\pi}.$$

- Lokaal maximum ($k = 2n + 1$):

$$x = \frac{3\pi}{4} + 2n\pi, \quad f(x) = +\frac{\sqrt{2}}{2} e^{\frac{3\pi}{4} + 2n\pi}.$$

References